CFD Analysis of the Propulsive Jet Effect on the Aerodynamic Coefficients of a Sounding Rocket

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Contents

1 Objectives and scope

This report investigates, for a selected flight condition of a sounding rocket, the effect of the propulsive jet on the mean aerodynamic coefficients predicted by CFD. The aim is not to provide a complete aerodynamic characterization of the vehicle over its whole flight envelope, but to quantify the error introduced when the exhaust jet is neglected and the nozzle exit is replaced by a closed surface.

The adopted method follows a comparative CFD workflow. First, a simplified but physically consistent model of the rocket and of the exhaust jet is defined. The numerical setup is then designed and checked through convergence criteria, domain-independence, mesh-independence, and near-wall validity checks, so that the comparison between the jet and no-jet configurations is based on numerically reliable mean quantities. The resulting coefficients are finally interpreted both globally and with specific attention to the base region, where the main aerodynamic differences are expected to arise.

More specifically, the report aims to:

- define a representative flight condition and the corresponding free-stream boundary conditions;
- model the powered and unpowered configurations in a consistent way;
- design the computational domain and mesh with adequate refinement in the wall region, wake, and base region;
- verify the numerical reliability of the simulations through convergence, domain-independence, mesh-independence, and near-wall checks;
- quantify the effect of jet modeling on the mean coefficients C_L, C_D, and C_M;
- identify the region mainly responsible for the observed aerodynamic differences and discuss their physical origin.

The scope of the document is intentionally limited to the implementation and verification of the CFD workflow required for this comparison. In particular, the report is not meant to provide a full aerodynamic database of the vehicle, a complete experimental validation of the absolute coefficients, or a high-fidelity simulation of internal nozzle flow, combustion chemistry, and fully unsteady plume dynamics. These aspects are simplified whenever they are not essential to the present objective.

The results should therefore be interpreted as a numerically supported engineering assessment of the aerodynamic impact of the propulsive jet under the selected operating condition.

2 Modeling choices

The present problem is a compressible external flow around a rocket, with an additional supersonic exhaust jet in the powered case. The flow is therefore described by the compressible Navier–Stokes equations, closed by an equation of state and solved numerically by CFD. Free-stream conditions are imposed on the outer boundaries, while the rocket surface is treated as a no-slip wall [10–12].

Since the objective of the study is the prediction of mean aerodynamic coefficients, a RANS approach was selected. Compared with DNS and LES, RANS does not resolve the full unsteady turbulent field, but models its effect on the mean flow at a much lower computational cost. This makes it more suitable for engineering applications in which average loads are of primary interest.

In the RANS framework, each flow variable is decomposed into a mean and a fluctuating part:

$$\varphi = \langle \varphi \rangle + \varphi' \tag{1}$$

and the averaged equations introduce additional unknown terms, mainly the Reynolds stresses. In eddy-viscosity models, these terms are closed through the Boussinesq hypothesis, which relates them to the mean strain rate by means of a turbulent viscosity [4].

Among the common eddy-viscosity models, the k-ω SST model was adopted. This model combines the good near-wall behavior of the k-ω formulation with the better free-stream robustness of the k-ε formulation. Its main advantage for the present work is the limiter on turbulent viscosity, which improves the behavior in separated flows and adverse pressure gradients [9]. In compact form:

$$\nu_t = \frac{a_1 k}{\max(a_1\omega,\ \|\langle \mathbf{S} \rangle\| F_2)} \tag{2}$$

The wall region is especially important because aerodynamic loads depend on both wall pressure and wall shear stress:

$$\mathbf{F} = \int_S (-p_w\,\mathbf{n} + \tau_w\,\mathbf{s})\ dS \tag{3}$$

A full wall-resolved treatment would require a very fine near-wall mesh, so a high-Re RANS approach with wall functions was used. In this case, the inner part of the boundary layer is modeled instead of being fully resolved. The relevant dimensionless quantities are:

$$u_\tau = \sqrt{\frac{\tau_w}{\rho}} \qquad u^+ = \frac{u_t}{u_\tau} \qquad y^+ = \frac{u_\tau y}{\nu} \tag{4}$$

Standard wall functions are generally intended for first-cell positions in the logarithmic region, typically:

$$y_1^+ \in [20, 400] \tag{5}$$

provided that the local flow is reasonably compatible with the assumptions of the wall law [9].

From the numerical point of view, the equations are solved with the finite-volume method on a discrete mesh. For steady simulations, the solver introduces a pseudo-time to iteratively reach the steady solution. In compact form:

$$\frac{\varphi_{n+1} - \varphi_n}{\Delta\tau} + f_\varphi(\boldsymbol{\varphi}_{n+1}) = 0 \tag{6}$$

where $\Delta\tau$ has no physical meaning and is used only to improve convergence. The associated pseudo-Courant number is:

$$pCFL = \frac{\|\mathbf{u}\|\Delta\tau}{\Delta x} \tag{7}$$

As in any finite-volume simulation, the accuracy of the solution also depends on mesh quality, especially in the reconstruction of convective and diffusive fluxes. For this reason, skewness and orthogonal quality were monitored during mesh generation.

Finally, a density-based solver was preferred over a pressure-based one, since the present problem includes both a compressible external flow and a supersonic jet. This formulation is generally better suited to flows in which pressure-wave propagation and strong compressibility effects are relevant.

3 Problem statement

The aim of this study is to quantify, for a selected flight condition of the rocket, how much the mean aerodynamic coefficients change when the exhaust jet is modeled or neglected.

In powered flight, the external flow is affected by the exhaust plume. However, including the jet in the CFD model requires additional assumptions and a more complex setup. For this reason, the nozzle exit is often replaced by a closed surface. The present work evaluates the error introduced by this simplification by comparing a jet case and a no-jet case.

3.1 Geometry

The analyzed vehicle is the *Nemesis* sounding rocket, developed by Aurora Rocketry in 2025. Its main dimensions are:

$$l = 2930 \text{ mm} \qquad d = 150 \text{ mm} \qquad m = 27.1 \text{ kg} \tag{8}$$

A clean external geometry was used in the CFD model. Secondary details such as screws, rail buttons, and surface roughness were removed, since their contribution is expected to be nearly the same in both configurations and therefore largely cancel out in the final comparison. This simplification also reduces meshing difficulty and computational cost.

The reference system is defined with origin at the rear end of the rocket axis, the x axis aligned with one fin, the y axis normal to that fin, and the z axis aligned with the incoming flow. The simplified geometry used in the simulations is shown in Fig. 1.

Figure 1: Simplified geometry used in the CFD simulations.

For post-processing, the rocket was divided into three main regions: fuselage, tail, and base. Here, the base includes both the nozzle-exit section and the surrounding base ring.

3.2 Output quantities

The selected flight condition includes a nonzero angle of attack, imposed by rotating the rocket about the x axis. Because the configuration remains symmetric with respect to the plane of attack, the aerodynamic loads were written as:

$$\mathbf{F} = (0, L, D) \qquad \mathbf{M}_{CG} = (M, 0, 0) \tag{9}$$

where the moment is computed about the rocket center of gravity, located on the longitudinal axis at:

$$d_{CG} = 1300 \text{ mm} \tag{10}$$

from the rear end.

The aerodynamic coefficients are defined as [3]:

$$C_L = \frac{L}{q_\infty S_t} \qquad C_D = \frac{D}{q_\infty S_t} \qquad C_M = \frac{M}{q_\infty S_t d} \tag{11}$$

with:

$$S_t = 1.767 \cdot 10^4 \text{ mm}^2 \qquad q_\infty = \frac{1}{2}\rho_\infty U_\infty^2 \tag{12}$$

The main output of the study is the relative difference between the mean coefficients obtained with and without jet modeling:

$$\Delta\overline{C} = \left| \frac{\overline{C}^{(\mathrm{j})} - \overline{C}^{(\mathrm{nj})}}{\overline{C}^{(\mathrm{j})}} \right| \pm \delta\overline{C} \qquad \delta\overline{C} = \delta\overline{C}_e + \delta\overline{C}_m \tag{13}$$

For consistency, the same rocket surface was used for force and moment integration in both cases. The nozzle-exit section itself was excluded from the aerodynamic integration, as shown in Fig. 2.

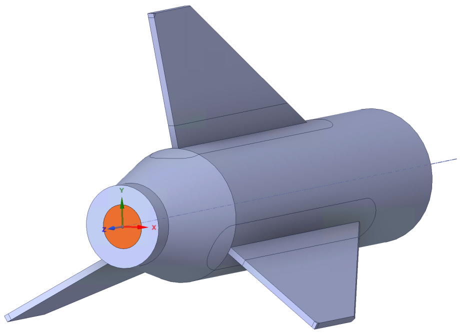

Figure 2: Nozzle-exit section excluded from the aerodynamic integration.

3.3 Reference flight condition and free-stream boundary conditions

Aerodynamic loads are evaluated by CFD simulations in ANSYS Fluent, using the flight velocity U and the corresponding atmospheric static conditions. The critical condition is identified by recalling that [3]:

$$L = qSC_L \qquad D = qSC_D \tag{14}$$

with [3]:

$$q = \frac{1}{2}\rho U^2 \tag{15}$$

Assuming a linear aerodynamic regime [3]:

$$C_L = C_{L_\alpha}\,\alpha \qquad C_D = C_{D_0} + k\,C_L^2 \tag{16}$$

lift scales with $q\,\alpha$, while drag depends on both q and α^2. Therefore:

$$L = qSC_{L_\alpha}\,\alpha \qquad D = qS\left(C_{D_0} + k\,C_{L_\alpha}^2\alpha^2\right) \tag{17}$$

Strictly speaking, the most critical aerodynamic condition for lift is the one that maximizes the quantity $q\,\alpha$, while drag is also affected by the baseline term C_{D_0}. In the present work, a conservative envelope condition is defined by combining the maximum angle of attack obtained from the trajectory analysis with the atmospheric state corresponding to the maximum dynamic pressure.

Maximum angle of attack

RocketPy trajectory analyses give the following maximum angle of attack:

$$|\alpha|_{\max} = 1.4^\circ \tag{18}$$

Maximum dynamic pressure

The dynamic pressure is maximized over the available ascent-trajectory data, from lift-off to apogee. The maximum value occurs at $t' = 4.996\,\mathrm{s}$, where

$$q_{\max} = 3.774 \cdot 10^4 \text{ Pa} \tag{19}$$

The corresponding atmospheric conditions, obtained from the ISA76 model, are [7]:

$$P = 9.139 \cdot 10^4 \text{ Pa} \qquad \rho = 1.127 \text{ kg/m}^3 \qquad T = 282.5 \text{ K} \tag{20}$$

while the flight condition adopted in the aerodynamic simulations is:

$$U = 258.8 \text{ m/s} \qquad M = 0.7680 \tag{21}$$

Turbulence was modeled with the RANS $k-\omega$ SST model. The inlet turbulence quantities were set in terms of turbulence intensity and viscosity ratio to the default values:

$$I = 0.05 \qquad VR = 10$$

3.4 Jet modeling

In the powered-flight case, the exhaust plume was modeled as an ideal compressible gas imposed at the nozzle-exit section. Since the chemical composition of the plume is not relevant to the present study, no combustion chemistry was included.

Because the exact nozzle geometry of the Cesaroni Pro75 8187M1545-P motor was not available, the throat and exit areas were taken from a similar Cesaroni Pro75 motor:

$$A^* = 451.6 \text{ mm}^2 \qquad A_e = 1903 \text{ mm}^2 \tag{22}$$

The exit conditions were estimated through an axisymmetric isentropic nozzle model. Using $\gamma_g = 1.2$, the area–Mach relation gives [14]:

$$\frac{A_e}{A^*} = \frac{1}{M_e}\left[\frac{2}{\gamma_g+1}\left(1+\frac{\gamma_g-1}{2}M_e^2\right)\right]^{\frac{\gamma_g+1}{2(\gamma_g-1)}} \qquad \Rightarrow \qquad M_e = 2.659 \tag{23}$$

Assuming chamber conditions:

$$T_c = 2800 \text{ K} \qquad p_c = 1.226 \cdot 10^6 \text{ Pa} \tag{24}$$

the exit temperature and pressure follow from the standard isentropic relations:

$$\frac{T_c}{T_e} = 1 + \frac{\gamma_g-1}{2}M_e^2 \qquad \Rightarrow \qquad T_e = 1640 \text{ K} \tag{25}$$

$$\frac{p_c}{p_e} = \left(1+\frac{\gamma_g-1}{2}M_e^2\right)^{\frac{\gamma_g}{\gamma_g-1}} \qquad \Rightarrow \qquad p_e = 4.955 \cdot 10^4 \text{ Pa} \tag{26}$$

Using $R_g = 360$ J/(kg K), the exit velocity was then obtained from:

$$M_e = \frac{U_e}{\sqrt{\gamma_g R_g T_e}} \qquad \Rightarrow \qquad U_e = 2238 \text{ m/s} \tag{27}$$

Since the rocket is simulated at nonzero angle of attack, the jet velocity was imposed in the wind-axis reference system as:

$$\mathbf{u}_e = (0, -U_e \sin\alpha, U_e \cos\alpha) \qquad \Rightarrow \qquad \mathbf{u}_e = (0, -54.68, 2237) \text{ m/s} \tag{28}$$

The inlet turbulence quantities were set in terms of turbulence intensity and turbulent length scale, where D_e is the exit diameter:

$$I_e = 0.05 \qquad l_c = 0.07\, D_e$$

The resulting pressure ratio:

$$\frac{p_e}{p_\infty} = 0.5422 \tag{29}$$

shows that the jet is overexpanded. A possible concern in this condition is the formation of a normal shock inside the nozzle, which would invalidate the isentropic assumption.

A simple quasi-one-dimensional check was therefore carried out. The Mach number at which a normal shock would adapt the nozzle flow to the ambient pressure was obtained from: [1]

$$\frac{p_\infty}{p_c} = \left[\frac{(\gamma_g+1)M_s^2}{2+(\gamma_g-1)M_s^2}\right]^{\frac{\gamma_g}{\gamma_g-1}} \left[\frac{\gamma_g+1}{2\gamma_g M_s^2-(\gamma_g-1)}\right]^{\frac{1}{\gamma_g-1}} \qquad \Rightarrow \qquad M_s = 3.852 \tag{30}$$

The corresponding nozzle area was then obtained from the area–Mach relation:

$$\frac{A_s}{A^*} = \frac{1}{M_s}\left[\frac{2}{\gamma_g+1}\left(1+\frac{\gamma_g-1}{2}M_s^2\right)\right]^{\frac{\gamma_g+1}{2(\gamma_g-1)}} \qquad \Rightarrow \qquad A_s = 1.034 \cdot 10^4 \text{ mm}^2 \tag{31}$$

Since:

$$A_s > A_e \tag{32}$$

a normal shock, if present, would form outside the nozzle. The isentropic model is therefore acceptable for the present purpose.

As a final consistency check, the thrust predicted from the imposed jet conditions was compared with the experimental motor thrust at the selected flight instant t'. The theoretical thrust was evaluated as [14]:

$$\|\mathbf{p}\| = p_c A^* \sqrt{\frac{2\gamma_g^2}{\gamma_g-1}\left(\frac{2}{\gamma_g+1}\right)^{\frac{\gamma_g+1}{\gamma_g-1}}\left[1-\left(\frac{p_e}{p_c}\right)^{\frac{\gamma_g-1}{\gamma_g}}\right]} + (p_e - p_\infty)A_e \tag{33}$$

which gives:

$$\|\mathbf{p}\| \approx 720 \text{ N} \tag{34}$$

This matches the experimental thrust at the same instant, so the adopted jet boundary conditions can be considered consistent with the physical behavior of the motor.

4 Preliminary mesh design

The mesh was designed to resolve the wall region, the base flow, and the wake with adequate accuracy, while keeping the computational cost acceptable.

4.1 Domain and surface-mesh strategy

Because of the slender and nearly axisymmetric shape of the rocket, the fluid domain was defined as a cylindrical enclosure. The first reference domain, denoted as e_1, was chosen as:

$$H_{\mathrm{ff}} = 30d = 4.5\ \mathrm{m} \qquad H_{\mathrm{in}} = 30d = 4.5\ \mathrm{m} \qquad H_{\mathrm{out}} = 90d = 13.5\ \mathrm{m} \tag{35}$$

where d is the rocket diameter.

The surface mesh was defined from local characteristic lengths, so that the discretization remained consistent with the different geometric scales of the rocket. Along the main fuselage, the reference length was the rocket diameter, which gave a surface size:

$$fs_1 = 7.500\ \mathrm{mm} \tag{36}$$

A local refinement was introduced at the nose tip, where the curvature is higher, reducing the surface size to:

$$fs_{\mathrm{tip}} = 1.875\ \mathrm{mm} \tag{37}$$

In the tail region, the relevant scale is the fin root chord rather than the fuselage diameter. The corresponding surface size was therefore set to:

$$fs_2 = 5.703\ \mathrm{mm} \tag{38}$$

A further refinement was introduced at the fin leading edge, where the local thickness is small, leading to:

$$fs_{\mathrm{LE}} = 3.140\ \mathrm{mm} \tag{39}$$

At the base and around the nozzle-exit section, the surface sizing was defined from the nozzle-exit diameter, giving:

$$fs_3 = 4.923\ \mathrm{mm} \tag{40}$$

This refinement was considered important because the main aerodynamic differences between the jet and no-jet cases were expected to develop in the base region. For the whole surface mesh, a smooth growth rate of 1.2 was used.

4.2 Local volumetric refinement

Two levels of volumetric refinement were introduced through bodies of influence. The first one, denoted as Near BOI, surrounds the rocket and controls the region immediately outside the wall. Its purpose is to provide a smooth transition from the prism layers to the volume mesh and to maintain adequate resolution in the regions where the boundary layer, base recirculation, and the initial part of the jet develop.

The local cell size inside the Near BOI was kept of the same order as the outer boundary-layer size, with smaller growth rates in the tail and base regions. In order to resolve the near wake more accurately, the Near BOI was also extended downstream by $10D_e$.

A second refinement region, denoted as Wake BOI, enclosed both the rocket and the Near BOI and controlled the wake and plume development farther downstream. In this region, progressively coarser target sizes of:

$$50\ \mathrm{mm}, \qquad 100\ \mathrm{mm}, \qquad 200\ \mathrm{mm} \tag{41}$$

were adopted, with growth rate 1.1.

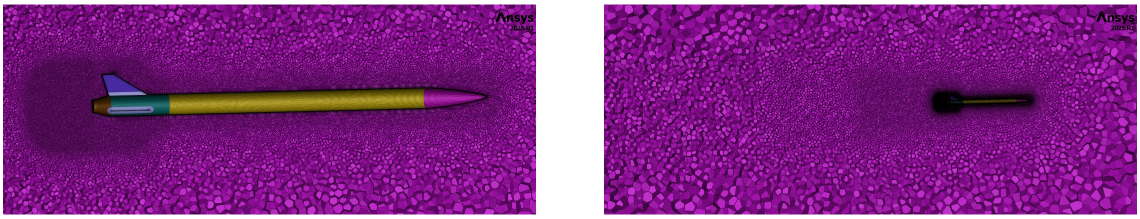

Figure 3: Local volumetric refinement: Near BOI (left) and Wake BOI (right).

4.3 Boundary-layer mesh

The near-wall region was resolved with uniform prism layers. Their total thickness T, first-layer thickness t_1, growth rate g, and number of layers N satisfy:

$$T = t_1 \frac{g^N - 1}{g - 1} \tag{42}$$

A first global boundary-layer definition was estimated from the free-stream Reynolds number:

$$Re_d = \frac{U_\infty d}{\nu_\infty} = 2.484 \cdot 10^6 \qquad Re_l = \frac{U_\infty l}{\nu_\infty} = 4.851 \cdot 10^7 \tag{43}$$

A turbulent flat-plate estimate then gave:

$$\delta_{99}(l) = \frac{0.37\, l}{Re_l^{1/5}} = 31.47 \text{ mm} \tag{44}$$

which was used as the reference total thickness:

$$T = 31.47 \text{ mm} \tag{45}$$

The first-cell location was designed for wall-function use, with target:

$$y_1^+ = 150 \tag{46}$$

Using the Prandtl–Schlichting correlation, the corresponding skin-friction coefficient, wall shear stress, and friction velocity were estimated as:

$$C_f = 2.059 \cdot 10^{-3} \qquad \tau_w = 77.71 \text{ Pa} \qquad u_\tau = 8.304 \text{ m/s} \tag{47}$$

which gave:

$$y = 0.2823 \text{ mm}, \qquad t_1 = 0.5646 \text{ mm} \tag{48}$$

A conservative choice of:

$$N = 20 \tag{49}$$

was adopted, leading to:

$$g = 1.098 \tag{50}$$

The global prism-layer definition was then corrected locally where it would have degraded mesh quality. At the nose tip, the prism layers were removed completely. At the outer fin edges, the total thickness was reduced to one quarter of the global value:

$$T_e = 7.868 \text{ mm} \tag{51}$$

which gave:

$$N_e = 10 \tag{52}$$

At the fin roots and in the base region, the total thickness was reduced to one half of the global value:

$$T_r = 15.74 \text{ mm} \tag{53}$$

which gave:

$$N_r = 15 \tag{54}$$

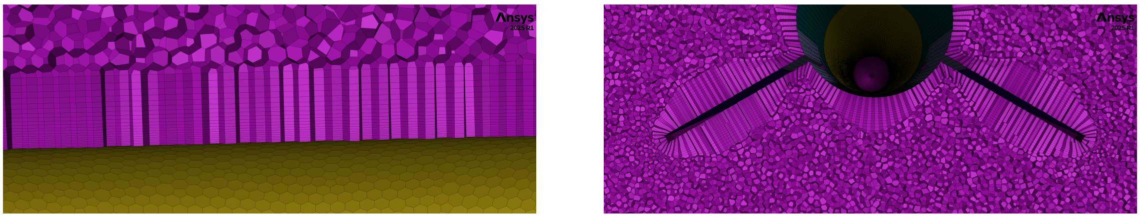

Figure 4: Global boundary layers (left) and local corrections (right).

In the far field, a target volume size of:

$$0.4 \text{ m} \tag{55}$$

was used with growth rate 1.2.

4.4 Mesh quality criteria

All generated meshes were required to satisfy:

$$\max(\text{Skw}) < 0.6 \qquad \min(\text{OQ}) > 0.4 \tag{56}$$

5 Numerical setup and convergence criteria

The simulations were carried out in ANSYS Fluent 2025 R1. A high-Re RANS approach with the SST k-ω model was adopted, since this model is generally robust for separated flows, especially in the base region.

Although the real flow is not strictly steady, the aim of the present work is the evaluation of mean aerodynamic coefficients. For this reason, all simulations were performed in steady-state mode, and the final coefficients were extracted from the converged oscillatory regime. The energy equation was kept active because the problem is compressible and includes a supersonic jet. A density-based solver was selected, with ideal-gas air, Sutherland's law for viscosity, kinetic-theory conductivity, and constant specific heats.

In steady RANS simulations, the aerodynamic coefficients rarely converge to perfectly flat values. Instead, they usually reach a bounded oscillatory regime, as shown in Fig. 5. Additional acceptance criteria were therefore introduced to ensure that the extracted mean values were numerically reliable.

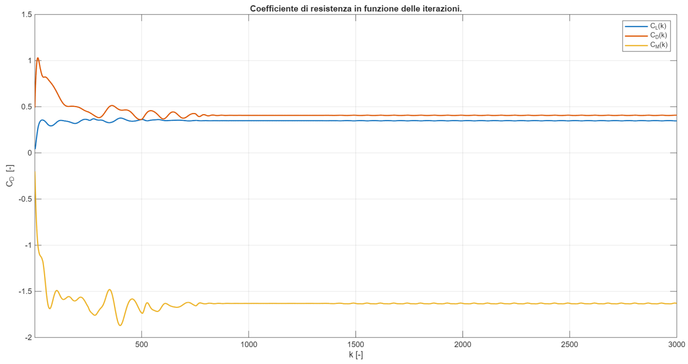

Figure 5: Typical convergence history of a steady simulation.

5.1 Acceptance criteria

A simulation was accepted only if the final regime satisfied the following conditions:

$$\varepsilon_{\overline{C}} < 0.1\% \qquad \varepsilon_{C_t} < 0.1\% \qquad \varepsilon_m < 0.1\% \qquad y^+_{1,\max} < 400 \qquad \eta_{F_b} > 99.9\% \tag{57}$$

Here, $\varepsilon_{\overline{C}}$ measures the sensitivity of the mean coefficient to the position of the averaging window, ε_{C_t} measures the residual slow drift, ε_m is the relative mass imbalance, $y^+_{1,\max}$ is the maximum first-cell wall coordinate over fuselage and tail, and η_{F_b} is the pressure contribution ratio at the base ring.

For a generic aerodynamic coefficient $C(n)$ sampled at iteration n, the final regime interval was defined as:

$$I_{\text{reg}} = [n_i, n_f] \tag{58}$$

The dominant oscillation period was estimated from the FFT of the final signal:

$$T_n = \left\lceil \frac{1}{f_n} \right\rceil \tag{59}$$

and the selected interval was required to contain at least ten periods:

$$\text{len}(I_{\text{reg}}) \geq 10\,T_n \tag{60}$$

A moving average over one period was then computed:

$$\overline{C}_{MA}(n) = \frac{1}{T_n} \sum_{i=n}^{n+T_n-1} C(i) \tag{61}$$

The relative spread of this moving average was used to define:

$$\varepsilon_{\overline{C}} = \frac{\max_n \overline{C}_{MA}(n) - \min_n \overline{C}_{MA}(n)}{|\overline{C}|} \tag{62}$$

where $\overline{C}$ is the mean value over the whole selected regime. A linear regression of $\overline{C}_{MA}(n)$:

$$\overline{C}_{MA}(n) \approx m_{\text{reg}}\, n + q_{\text{reg}} \tag{63}$$

was used to define the relative slow trend:

$$\varepsilon_{C_t} = \frac{|m_{\text{reg}}|\, \text{len}(I_{MA})}{|\overline{C}|} \tag{64}$$

Residual monitoring was complemented by a global mass-conservation check. For steady simulations, the relative mass imbalance was defined as:

$$\varepsilon_m = \max_{n \in I_{\text{reg}}} \left| \frac{\dot{m}_{\text{walls}}(n)}{\dot{m}_{\text{in}}(n)} \right| \tag{65}$$

where $\dot{m}_{\text{walls}}$ is the total mass flow crossing the domain boundaries, including the jet inlet in the powered case.

The near-wall validity check was based on the first-cell wall coordinate. Over fuselage and tail:

$$y^+_{1_{\max}} = \max_{\substack{\mathbf{x} \in \text{f+t} \\ n \in I_{\text{reg}}}} y^+_1(\mathbf{x}, n) < 400 \tag{66}$$

At the base ring, where separation and jet effects make a direct y^+ criterion less meaningful, the load was decomposed into pressure and shear contributions:

$$\mathbf{F}_b = \mathbf{F}_{b_p} + \mathbf{F}_{b_\tau} \tag{67}$$

and the pressure contribution ratio was defined as:

$$\eta_{F_b} = \max_{n \in I_{\text{reg}}} \frac{\|\mathbf{F}_{b_p}(n)\|}{\|\mathbf{F}_{b_p}(n)\| + \|\mathbf{F}_{b_\tau}(n)\|} \tag{68}$$

This condition ensures that any possible inaccuracy in the base shear stress has a negligible effect on the total base load.

5.2 Pseudo-Courant strategy

The pseudo-Courant number was initially set to:

$$pCFL = 5 \tag{69}$$

to stabilize the solver in the early iterations. After residual reduction and the onset of a bounded oscillatory regime, it was progressively increased through:

$$5 \to 10 \to 20 \to 50 \to 100 \to 200 \to 500 \tag{70}$$

Once the final oscillatory regime was reached, the pseudo-Courant number was set back to the standardized value:

$$pCFL = 200 \tag{71}$$

and kept fixed during data collection, so that all mean coefficients were extracted under the same numerical conditions.

6 Domain-independence study

The outer boundaries of the fluid domain must be sufficiently far from the rocket to avoid artificial interference with the flow field, while still keeping the computational cost under control. A domain-independence study was therefore carried out.

6.1 Validation criterion

The final output of the study is affected by both domain and mesh uncertainty:

$$\delta\overline{C} = \delta\overline{C}_e + \delta\overline{C}_m \tag{72}$$

where $\delta\overline{C}_e$ is the contribution associated with the computational domain.

For two consecutive domains, e_i and e_{i+1}, the relative jet effect is defined as:

$$\Delta\overline{C}_{e_i} = \frac{\overline{C}^{(\mathrm{j})}_{e_i} - \overline{C}^{(\mathrm{nj})}_{e_i}}{\overline{C}^{(\mathrm{j})}_{e_i}} \qquad \Delta\overline{C}_{e_{i+1}} = \frac{\overline{C}^{(\mathrm{j})}_{e_{i+1}} - \overline{C}^{(\mathrm{nj})}_{e_{i+1}}}{\overline{C}^{(\mathrm{j})}_{e_{i+1}}} \tag{73}$$

The domain-related uncertainty associated with e_i is then estimated as:

$$\delta\overline{C}_{e_i} = \left|\Delta\overline{C}_{e_{i+1}} - \Delta\overline{C}_{e_i}\right| \tag{74}$$

A domain is considered acceptable if:

$$\delta\overline{C}_{e_i} < 1\% \tag{75}$$

for all the mean aerodynamic coefficients of interest. Successive domains were generated by uniformly scaling the previous one by a factor of 1.5.

An additional check was introduced on the selected domain to verify the absence of outlet backflow. Defining:

$$w_{\mathrm{out}} = \mathbf{u} \cdot \mathbf{n}_{\mathrm{out}} \tag{76}$$

the minimum outlet-normal velocity over the whole outlet section and over the selected regime interval was required to satisfy:

$$w_{\mathrm{out_{min}}} = \min_{\substack{\mathbf{x}\in A_{\mathrm{out}} \\ n\in I_{\mathrm{reg}}}} w_{\mathrm{out}}(\mathbf{x}, n) > 0 \tag{77}$$

6.2 Results

The mean aerodynamic coefficients obtained on the three tested domains are reported in Tab. 1.

Table 1: Mean aerodynamic coefficients in e_1, e_2, and e_3.

Sim.	$\overline{C}_L$		$\overline{C}_D$		$\overline{C}_M$	
	j	nj	j	nj	j	nj
e_1	0.3463	0.3488	0.4662	0.4067	−1.625	−1.632
e_2	0.3402	0.3452	0.4620	0.4051	−1.5800	−1.608
e_3	0.3393	0.3452	0.4662	0.4070	−1.575	−1.610

The corresponding relative variations between consecutive domains are reported in Tab. 2.

Table 2: Relative variations between consecutive domains.

Sim.	$\delta\overline{C}_L$ [%]	$\delta\overline{C}_D$ [%]	$\delta\overline{C}_M$ [%]
$e_{1\to 2}$	0.7554 ✓	0.4405 ✓	1.275 ✗
$e_{2\to 3}$	0.2636 ✓	0.3664 ✓	0.4606 ✓

Domain e_1 cannot yet be considered independent, since the variation in $\overline{C}_M$ between e_1 and e_2 exceeds the imposed tolerance. The comparison between e_2 and e_3 shows instead that all the considered coefficients satisfy the 1% criterion. Domain e_2 was therefore selected for the subsequent mesh-independence study.

The outlet backflow check on e_2 gave:

$$w^{(\mathrm{j})}_{\mathrm{out}_{\min}} = 258.7\,\mathrm{m/s}\ \checkmark \qquad w^{(\mathrm{nj})}_{\mathrm{out}_{\min}} = 257.9\,\mathrm{m/s}\ \checkmark \tag{78}$$

showing that the flow remains outward over the whole outlet section in both cases.

The relative variations are also shown in Fig. 6.

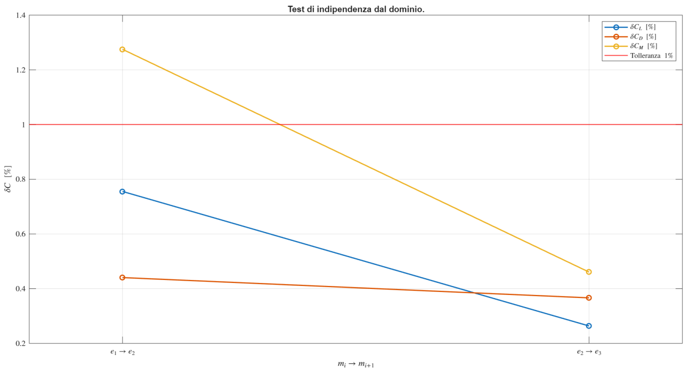

Figure 6: Relative variations in the domain-independence study.

7 Mesh-independence study

The mesh controls how accurately the flow field and the resulting aerodynamic loads are resolved. In particular, insufficient resolution increases numerical diffusion and may prevent the correct capture of strong local gradients, especially in the boundary layer, wake, and jet region. A mesh-independence study was therefore carried out to verify that the predicted mean aerodynamic coefficients are sufficiently insensitive to further refinement.

7.1 Validation criterion

As for the domain study, the total uncertainty on the final result is written as

$$\delta\overline{C} = \delta\overline{C}_e + \delta\overline{C}_m \tag{79}$$

where $\delta\overline{C}_m$ is the contribution associated with the mesh.

For two meshes of increasing refinement, m_i and m_{i+1}, the relative jet effect is defined as:

$$\Delta\overline{C}_{m_i} = \frac{\overline{C}^{(\mathrm{j})}_{m_i} - \overline{C}^{(\mathrm{nj})}_{m_i}}{\overline{C}^{(\mathrm{j})}_{m_i}} \qquad \Delta\overline{C}_{m_{i+1}} = \frac{\overline{C}^{(\mathrm{j})}_{m_{i+1}} - \overline{C}^{(\mathrm{nj})}_{m_{i+1}}}{\overline{C}^{(\mathrm{j})}_{m_{i+1}}} \tag{80}$$

The mesh-related uncertainty associated with m_i is then estimated as:

$$\delta\overline{C}_{m_i} = \left|\Delta\overline{C}_{m_{i+1}} - \Delta\overline{C}_{m_i}\right| \tag{81}$$

A mesh is considered acceptable if:

$$\delta\overline{C}_{m_i} < 1\% \tag{82}$$

for all the mean aerodynamic coefficients of interest.

Since the study focuses on the effect of the jet, the initial refinement strategy was concentrated mainly in the rear part of the rocket, especially in the tail and base regions inside the Near BOI. Successive meshes were first generated by reducing the local cell sizes in these regions by a factor of 1.25.

7.2 Results

Because mesh m_1 coincides with the one already used on the validated domain e_2, its results were already available. The mean aerodynamic coefficients obtained on meshes m_1, m_2, and m_3 are reported in Tab. 3.

Table 3: Mean aerodynamic coefficients on meshes m_1, m_2, and m_3.

Sim.	$\overline{C}_L$		$\overline{C}_D$		$\overline{C}_M$	
	j	nj	j	nj	j	nj
m_1	0.3402	0.3452	0.4620	0.4051	−1.580	−1.608
m_2	0.3419	0.3457	0.4467	0.3985	−1.589	−1.608
m_3	0.3394	0.3434	0.4379	0.3989	−1.568	−1.591

The corresponding relative variations are reported in Tab. 4.

Table 4: Relative variations between consecutive meshes.

Sim.	$\delta\overline{C}_L$ [%]	$\delta\overline{C}_D$ [%]	$\delta\overline{C}_M$ [%]
$m_{1\to2}$	0.3452 ✓	1.517 ✗	0.5124 ✓
$m_{2\to3}$	0.07395 ✓	1.878 ✗	0.2387 ✓

Neither m_1 nor m_2 can be considered mesh-independent, since $\delta\overline{C}_D$ remains above the imposed tolerance. Moreover, the refinement from m_2 to m_3 does not produce an appreciable improvement in the drag-related uncertainty. This indicated that a further broad refinement of the whole tail region would likely be inefficient, and that the residual sensitivity had to be localized more precisely.

To identify the source of the remaining mesh dependence, the variation of $\overline{C}_D$ between m_2 and m_3 was examined separately for the jet and no-jet cases:

$$\overline{C}^{(\mathrm{j})}_{D_{m_3}} - \overline{C}^{(\mathrm{j})}_{D_{m_2}} = -0.008770 \qquad \overline{C}^{(\mathrm{nj})}_{D_{m_3}} - \overline{C}^{(\mathrm{nj})}_{D_{m_2}} = 4.027 \cdot 10^{-4} \tag{83}$$

The residual sensitivity is therefore almost entirely dominated by the jet case. In relative terms:

$$\eta_j = \frac{\left|\overline{C}^{(\mathrm{j})}_{D_{m_3}} - \overline{C}^{(\mathrm{j})}_{D_{m_2}}\right|}{\left|\overline{C}^{(\mathrm{j})}_{D_{m_3}} - \overline{C}^{(\mathrm{j})}_{D_{m_2}}\right| + \left|\overline{C}^{(\mathrm{nj})}_{D_{m_3}} - \overline{C}^{(\mathrm{nj})}_{D_{m_2}}\right|} \qquad \Rightarrow \qquad \eta_j = 95.61\% \tag{84}$$

The drag coefficient in the jet case was then decomposed into a base contribution and a contribution from the rest of the rocket:

$$\overline{C}^{(\mathrm{j})}_{D} = \overline{C}^{(\mathrm{j})}_{D_b} + \overline{C}^{(\mathrm{j})}_{D_r} \tag{85}$$

Between m_2 and m_3, the corresponding variations are:

$$\overline{C}^{(\mathrm{j})}_{D_{bm_3}} - \overline{C}^{(\mathrm{j})}_{D_{bm_2}} = -0.008674 \qquad \overline{C}^{(\mathrm{j})}_{D_{rm_3}} - \overline{C}^{(\mathrm{j})}_{D_{rm_2}} = -9.622 \cdot 10^{-5} \tag{86}$$

Therefore:

$$\eta_{b,j} = \frac{\left|\overline{C}^{(\mathrm{j})}_{D_{bm_3}} - \overline{C}^{(\mathrm{j})}_{D_{bm_2}}\right|}{\left|\overline{C}^{(\mathrm{j})}_{D_{bm_3}} - \overline{C}^{(\mathrm{j})}_{D_{bm_2}}\right| + \left|\overline{C}^{(\mathrm{j})}_{D_{bm_3}} - \overline{C}^{(\mathrm{j})}_{D_{bm_2}}\right|} \qquad \Rightarrow \qquad \eta_{b,j} = 98.90\% \tag{87}$$

This result shows that the remaining mesh sensitivity is almost entirely concentrated on the base ring in the jet case. A further broad refinement of the whole tail region would therefore have increased the computational cost without a proportional benefit. For this reason, a second mesh family was generated with a more localized refinement near the base.

The resulting mean aerodynamic coefficients are reported in Tab. 5.

Table 5: Mean aerodynamic coefficients on meshes m_3, m_4, and m_5.

Sim.	$\overline{C}_L$		$\overline{C}_D$		$\overline{C}_M$	
	j	nj	j	nj	j	nj
m_3	0.3394	0.3434	0.4379	0.3989	−1.568	−1.591
m_4	0.3407	0.3435	0.4313	0.3982	−1.577	−1.591
m_5	0.3414	0.3442	0.4269	0.3978	−1.581	−1.596

The corresponding relative variations are reported in Tab. 6.

Table 6: Relative variations between consecutive meshes in the localized-refinement study.

Sim.	$\delta\overline{C}_L$ [%]	$\delta\overline{C}_D$ [%]	$\delta\overline{C}_M$ [%]
$m_{3\to4}$	0.3439 ✓	1.233 ✗	0.5465 ✓
$m_{4\to5}$	0.02206 ✓	0.8733 ✓	0.01060 ✓

The localized refinement effectively reduced $\delta\overline{C}_D$ below the imposed threshold in the $m_4 \to m_5$ step. Mesh m_4 was therefore selected as the validated mesh for the subsequent simulations.

The relative variations are also shown in Fig. 7.

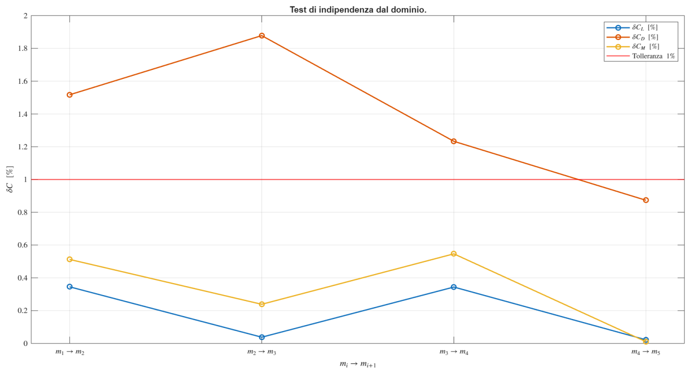

Figure 7: Relative variations in the mesh-independence study.

8 Results

8.1 Final coefficient differences

The final output of the study is expressed as:

$$\Delta\overline{C} = \left|\frac{\overline{C}^{(\mathrm{j})} - \overline{C}^{(\mathrm{nj})}}{\overline{C}^{(\mathrm{j})}}\right| \pm \delta\overline{C} \qquad [\%] \tag{88}$$

where (j) and (nj) denote the jet and no-jet cases.

The nominal values were computed from mesh m_4, selected as the validated mesh after comparison with m_5:

$$\Delta\overline{C}_{L_n} = 0.8170\%, \qquad \Delta\overline{C}_{D_n} = 6.816\%, \qquad \Delta\overline{C}_{M_n} = 0.9350\% \tag{89}$$

The total uncertainty was taken as:

$$\delta\overline{C} = \delta\overline{C}_e + \delta\overline{C}_m \tag{90}$$

which gives:

$$\delta\overline{C}_L = 0.2856\% \qquad \delta\overline{C}_D = 1.240\% \qquad \delta\overline{C}_M = 0.4712\% \tag{91}$$

The final results are therefore:

$$\begin{aligned} \Delta\overline{C}_L &= 0.8170\% \pm 0.2856\% \\ \Delta\overline{C}_D &= 6.816\% \pm 1.240\% \\ \Delta\overline{C}_M &= 0.9350\% \pm 0.4712\% \end{aligned} \tag{92}$$

8.2 Result interpretation

For $\overline{C}_L$ and $\overline{C}_M$, the differences between the jet and no-jet cases are small. Within the estimated uncertainty, the two configurations therefore provide comparable mean lift and pitching-moment coefficients.

The behavior of $\overline{C}_D$ is different. On the validated mesh m_4:

$$\overline{C}_D^{(\mathrm{j})} = 0.4313 \qquad \overline{C}_D^{(\mathrm{nj})} = 0.3982 \tag{93}$$

so that:

$$\overline{C}_D^{(\mathrm{j})} - \overline{C}_D^{(\mathrm{nj})} = 0.03317 \tag{94}$$

corresponding to a drag increase of about 6.8% in the jet case.

To identify the origin of this difference, the drag contribution of the base ring was separated from that of the rest of the rocket:

$$\overline{C}_{D_b}^{(\mathrm{j})} - \overline{C}_{D_b}^{(\mathrm{nj})} = 0.03493 \qquad \overline{C}_{D_r}^{(\mathrm{j})} - \overline{C}_{D_r}^{(\mathrm{nj})} = -1.760 \cdot 10^{-3} \tag{95}$$

Hence:

$$\eta_b = \frac{\left|\overline{C}_{D_b}^{(\mathrm{j})} - \overline{C}_{D_b}^{(\mathrm{nj})}\right|}{\left|\overline{C}_{D_b}^{(\mathrm{j})} - \overline{C}_{D_b}^{(\mathrm{nj})}\right| + \left|\overline{C}_{D_r}^{(\mathrm{j})} - \overline{C}_{D_r}^{(\mathrm{nj})}\right|} \qquad \Rightarrow \qquad \eta_b = 95.20\% \tag{96}$$

The drag variation is therefore almost entirely dominated by the base ring. The main effect of the jet is not a general increase in drag over the whole rocket, but a local modification of the aerodynamic contribution of the base region.

To clarify the physical origin of this behavior, the area-averaged pressure coefficient on the base ring was compared in the two cases:

$$\overline{C}_{p_b}^{(\mathrm{j})} = \frac{\overline{p}_b^{(\mathrm{j})} - p_\infty}{q_\infty} \qquad \overline{C}_{p_b}^{(\mathrm{nj})} = \frac{\overline{p}_b^{(\mathrm{nj})} - p_\infty}{q_\infty} \tag{97}$$

The resulting values are:

$$\overline{C}_{p_b}^{(\mathrm{j})} = -0.05978 \qquad \overline{C}_{p_b}^{(\mathrm{nj})} = 0.04253 \tag{98}$$

In the jet case, the base ring is therefore under suction, which is consistent with the increase in drag. In the no-jet case, instead, the mean pressure on the base ring is slightly above the free-stream value, producing a weak thrust-like contribution, that is, a negative local base drag.

A qualitative comparison of the $\overline{C}_p$ distribution is shown in Fig. 8.

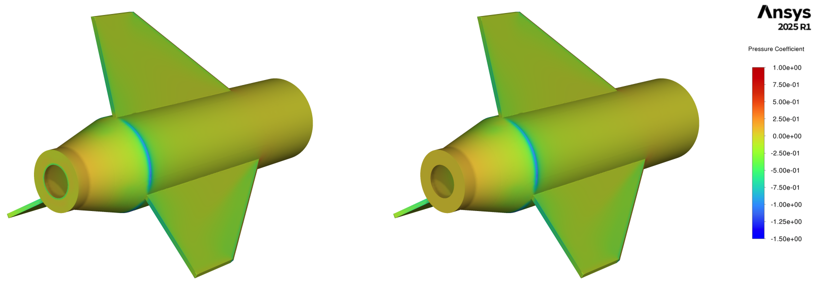

Figure 8: Mean pressure-coefficient distribution. Jet case on the left, no-jet case on the right.

The sign obtained in the no-jet case should not be interpreted automatically as a numerical artifact. NASA reports show that, for blunt-base configurations in subsonic flow, base drag can become negative, and that the Mach number above which this contribution becomes positive again is of the order of $M \approx 1.1$. Since the present flight condition is at about $M = 0.77$, the no-jet result is consistent with experimental evidence [13]. The same trend is also consistent with the discussion reported by Buresti for axisymmetric bodies with blunt base and boat-tail, where the boat-tail can increase base pressure up to the point of producing a thrust-like base contribution [2].

In the jet case, the pressure on the base ring decreases and its contribution becomes drag-producing. This is consistent with NASA experimental results showing that, under overexpanded conditions, a jet exiting at a pressure lower than the ambient one must readapt to the external field. This interaction can intensify the suction in the recirculation region behind the base, reduce the average rear pressure, and increase the drag contribution of the base region [5, 6].

In the present study, the jet is overexpanded, since:

$$\frac{p_e}{p_\infty} \approx 0.54 \tag{99}$$

so the observed drag increase is physically consistent with the considered operating condition.

It should finally be noted that the effect of the jet on drag is not universal, but depends on the operating regime. At higher pressure ratios, experimental results show that the pressure on the base ring may rise rapidly and even exceed the no-jet value, leading to the opposite effect, namely a drag reduction.

9 Appendix A: Flow-field visualization

9.1 Velocity magnitude

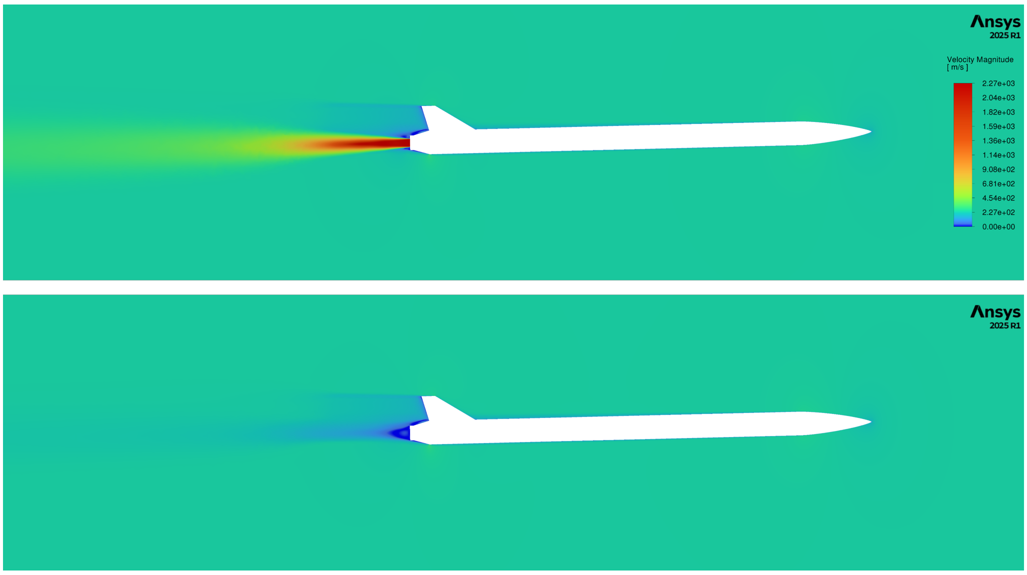

Figure 9: Global contours of velocity magnitude. Jet case on top, no-jet case on bottom.

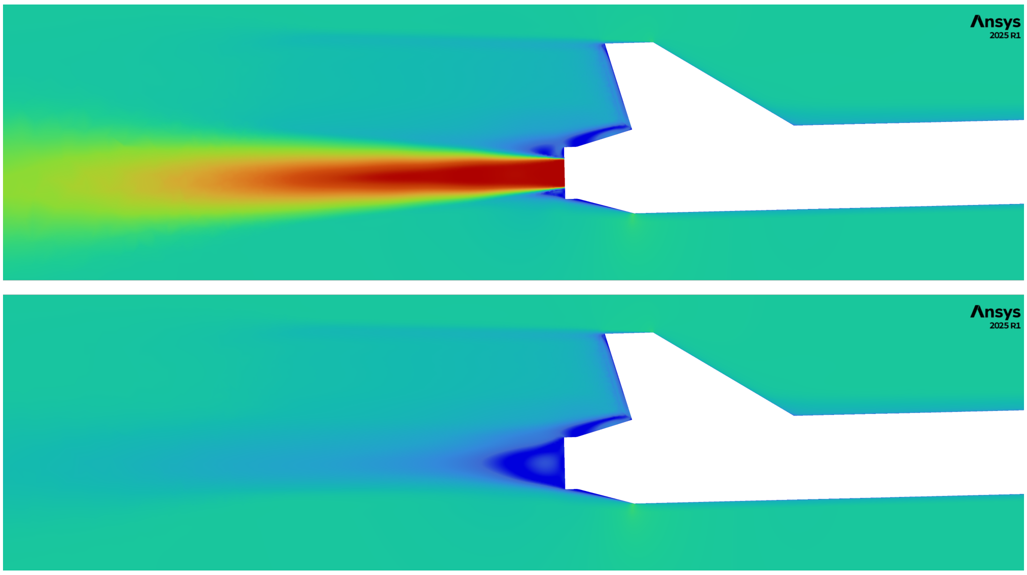

Figure 10: Local contours of velocity magnitude near the base region. Jet case on top, no-jet case on bottom.

9.2 Static pressure

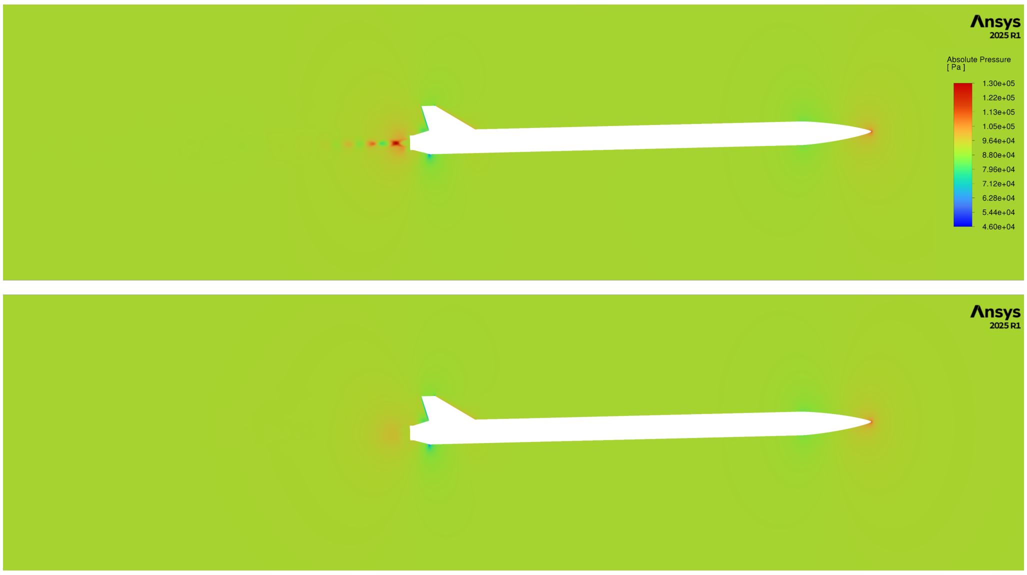

Figure 11: Global static-pressure contours. Jet case on top, no-jet case on bottom.

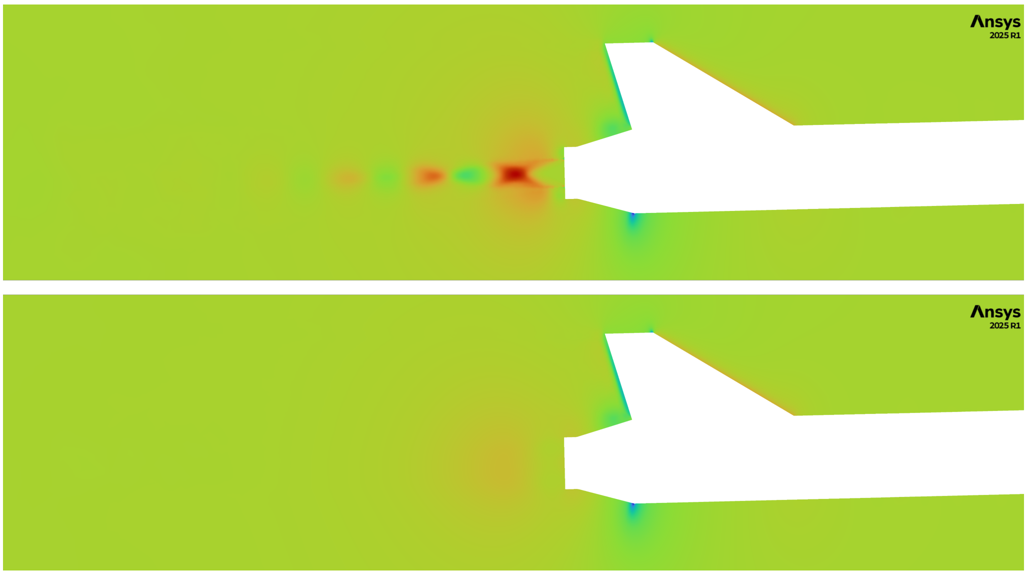

Figure 12: Local static-pressure contours near the base region. Jet case on top, no-jet case on bottom.

In the jet case, besides the lower pressure near the base, the plume also shows a lobed structure that is consistent with Mach diamonds. This behavior is expected for an overexpanded jet: the exhaust leaves the nozzle at a pressure lower than the ambient one and is therefore compressed by the surrounding flow. Because this compression overshoots the equilibrium condition, it is followed by re-expansion, and the process repeats in an oscillatory way while gradually damping downstream [8].

10 Appendix B: Numerical checks of the simulations

10.1 Mesh-quality verification.

Table 7: Mesh-quality parameters.

Sim.	Skw	OQ
e_1	0.52	0.48
e_2	0.52	0.48
e_3	0.59	0.41
m_2	0.56	0.44
m_3	0.57	0.43
m_4	0.51	0.49
m_5	0.52	0.48

10.2 Statistical-stationarity verification of the final regime

10.2.1 Verification of independence from the window position

Table 8: Relative errors associated with the window position.

Sim.	$\varepsilon_{\overline{C}_L}$ [%]		$\varepsilon_{\overline{C}_D}$ [%]		$\varepsilon_{\overline{C}_M}$ [%]	
	j	nj	j	nj	j	nj
e_1	$1.3 \cdot 10^{-3}$	$4.9 \cdot 10^{-3}$	$1.7 \cdot 10^{-2}$	$5.8 \cdot 10^{-3}$	$1.9 \cdot 10^{-3}$	$3.9 \cdot 10^{-3}$
e_2	$1.5 \cdot 10^{-2}$	$1.5 \cdot 10^{-4}$	$4.5 \cdot 10^{-3}$	$1.7 \cdot 10^{-4}$	$2.0 \cdot 10^{-2}$	$7.4 \cdot 10^{-2}$
e_3	$2.4 \cdot 10^{-2}$	$4.0 \cdot 10^{-3}$	$6.6 \cdot 10^{-3}$	$4.6 \cdot 10^{-3}$	$2.9 \cdot 10^{-2}$	$5.2 \cdot 10^{-3}$
m_2	$2.0 \cdot 10^{-2}$	$7.9 \cdot 10^{-3}$	$2.8 \cdot 10^{-3}$	$1.1 \cdot 10^{-2}$	$2.6 \cdot 10^{-2}$	$5.3 \cdot 10^{-3}$
m_3	$8.3 \cdot 10^{-3}$	$8.1 \cdot 10^{-3}$	$3.0 \cdot 10^{-3}$	$8.5 \cdot 10^{-3}$	$1.0 \cdot 10^{-2}$	$1.6 \cdot 10^{-2}$
m_4	$9.9 \cdot 10^{-2}$	$5.2 \cdot 10^{-2}$	$1.9 \cdot 10^{-2}$	$1.6 \cdot 10^{-2}$	$8.5 \cdot 10^{-2}$	$8.9 \cdot 10^{-2}$
m_5	$3.3 \cdot 10^{-2}$	$3.4 \cdot 10^{-2}$	$7.3 \cdot 10^{-3}$	$1.2 \cdot 10^{-2}$	$5.8 \cdot 10^{-2}$	$5.8 \cdot 10^{-2}$

10.2.2 Verification of the absence of slow drift

Table 9: Relative errors associated with slow drift.

Sim.	$\varepsilon_{C_{L_t}}$ [%]		$\varepsilon_{C_{D_t}}$ [%]		$\varepsilon_{C_{M_t}}$ [%]	
	j	nj	j	nj	j	nj
e_1	$2.1 \cdot 10^{-4}$	$8.5 \cdot 10^{-5}$	$2.1 \cdot 10^{-4}$	$5.5 \cdot 10^{-5}$	$3.7 \cdot 10^{-4}$	$2.5 \cdot 10^{-5}$
e_2	$2.7 \cdot 10^{-4}$	$3.9 \cdot 10^{-5}$	$7.5 \cdot 10^{-5}$	$3.1 \cdot 10^{-5}$	$6.0 \cdot 10^{-4}$	$3.1 \cdot 10^{-4}$
e_3	$1.2 \cdot 10^{-3}$	$5.6 \cdot 10^{-4}$	$4.6 \cdot 10^{-4}$	$6.0 \cdot 10^{-4}$	$3.0 \cdot 10^{-3}$	$2.2 \cdot 10^{-3}$
m_2	$5.7 \cdot 10^{-4}$	$9.1 \cdot 10^{-5}$	$3.9 \cdot 10^{-4}$	$1.2 \cdot 10^{-4}$	$8.7 \cdot 10^{-4}$	$1.8 \cdot 10^{-4}$
m_3	$5.0 \cdot 10^{-4}$	$4.2 \cdot 10^{-3}$	$4.6 \cdot 10^{-4}$	$4.4 \cdot 10^{-3}$	$2.7 \cdot 10^{-3}$	$1.4 \cdot 10^{-2}$
m_4	$1.8 \cdot 10^{-2}$	$5.6 \cdot 10^{-4}$	$2.4 \cdot 10^{-4}$	$6.1 \cdot 10^{-3}$	$2.5 \cdot 10^{-2}$	$1.1 \cdot 10^{-2}$
m_5	$1.5 \cdot 10^{-2}$	$1.1 \cdot 10^{-2}$	$2.8 \cdot 10^{-3}$	$3.0 \cdot 10^{-3}$	$2.0 \cdot 10^{-2}$	$7.7 \cdot 10^{-3}$

10.3 Continuity-equation convergence check

Table 10: Relative errors associated with mass imbalance.

Sim.	ε_m [%]	
	g	ng
e_1	$3.9 \cdot 10^{-3}$	$1.5 \cdot 10^{-4}$
e_2	$1.1 \cdot 10^{-4}$	$1.7 \cdot 10^{-4}$
e_3	$2.3 \cdot 10^{-4}$	$1.0 \cdot 10^{-4}$
m_2	$1.8 \cdot 10^{-3}$	$6.8 \cdot 10^{-5}$
m_3	$2.0 \cdot 10^{-3}$	$4.9 \cdot 10^{-5}$
m_4	$2.0 \cdot 10^{-3}$	$4.6 \cdot 10^{-5}$
m_5	$2.0 \cdot 10^{-3}$	$1.2 \cdot 10^{-4}$

10.4 Validity check of y_1^+

Table 11: Maximum values of y_1^+ and of the normal contribution at the base.

Sim.	$y^+_{1_{\max}}$		η_{F_b} [%]	
	j	nj	j	nj
e_1	194	194	100	100
e_2	193	193	100	100
e_3	197	196	100	100
m_2	193	193	100	100
m_3	194	194	100	100
m_4	194	194	100	100
m_5	194	194	100	100

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